

Generalized Bayes: the route back to standard Bayes I

The inference engine FedGVI federates is the generalized (Gibbs) posterior (Bissiri et al. 2016; Jiang and Tanner 2008; Jewson et al. 2018; Knoblauch et al. 2022), which trades the likelihood for a loss L and the KL regularizer for a general divergence $\mathcal{D}$:

$$q_n^*(s) = \arg \min_{q_n} \mathbb{E}_{q_n} \left[\sum_i L(s; o_i) \right] + \frac{1}{\tau} \mathcal{D}(q_n \parallel \pi_0), \quad (1)$$

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with prior π_0 , learning rate τ , and regularizing divergence $\mathcal{D}$. The learning rate is part of the inferential specification, not a cosmetic constant; coarsened-posterior and safe-Bayes work show why calibration of that temperature matters under misspecification (Miller and Dunson 2018; Grünwald 2012), where ordinary Bayes concentrates around a KL pseudo-truth rather than literal truth when the model family is wrong (Kleijn and Vaart 2012). We name the object eq. 1 defines.

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Definition (Generalized-(Gibbs)-Bayes posterior)

For a loss L , prior π_0 , learning rate $\tau > 0$, and divergence $\mathcal{D}$, the generalized-Bayes posterior is the minimizer q_n^* of (1). For $\mathcal{D} = \text{KL}$ the minimizer is the tempered softmax

$$q_n^*(s) \propto \pi_0(s) \exp\left(-\tau \sum_i L(s; o_i)\right),$$

implemented in `generalized_bayes.generalized_posterior`.

The tempered softmax of eq. 1, stated in the definition above, is not an approximation: it is the exact closed-form minimizer of eq. 1 when the regularizer is the KL divergence, because the categorical support is finite and the objective is strictly convex in q . The recovery to standard Bayes follows by choosing the loss. With $L = \text{NLL}$, $\text{NLL}(p, o) = -\log p(o)$, the exponential in that tempered softmax becomes a product of likelihoods and the minimizer is

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exactly standard Bayes; eq. 6 in sec. 5.8 states that corner, and Corollary 6 there pins it to the closed-form prior-times-likelihood product. The largest observed discrepancy between `generalized_posterior` in this regime and the analytic Bayes posterior is $5.55\text{e-}17$, reported in sec. 7.1 — exact to machine precision (a maximum deviation of about one ULP), not merely close.

FedGVI computes each client update against a *cavity* rather than the full posterior, so a contributing agent does not double-count its own previous message. We name that operation.

Definition (Cavity / PVI factor update)

The cavity removes agent n 's factor from the colony posterior in natural-parameter (log) space,

$$q_{-n}(s) = \frac{q(s)/t_n(s)}{\sum_{s'} q(s')/t_n(s')} = \text{softmax}(\log q(s) - \log t_n(s)),$$

where the final expression makes the normalization explicit; the partitioned-variational-inference (PVI) update re-multiplies a refreshed factor onto the cavity of (2). Taking a cavity and re-multiplying the original site factor restores the global posterior

$$q(s) = \frac{q_{-n}(s)t_n(s)}{\sum_{s'} q_{-n}(s')t_n(s')}, \quad (2)$$

with the original site factor, the recombination identity, the property `generalized_bayes.cavity` and `generalized_bayes.update_factor` satisfy.

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The numbered recombination identity is eq. 2.

The cavity of eq. 2 is the discrete analogue of the expectation-propagation / partitioned-VI cavity used outside active inference (Ashman et al. 2022; Bui et al. 2018), imported here so that the per-agent generalized-Bayes update of eq. 1 is computed against the colony belief with the agent's own contribution removed — exactly the sensory-attenuation discipline the belief-sharing round of sec. 5.9 requires. What remains unspecified in eq. 1 are its two ingredients — the divergence $\mathcal{D}$ and the loss L — whose robust members and standard-Bayes limits sec. 5.6 develops next; the aggregation identity (sec. 5.8) then federates the resulting per-agent posteriors.

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The authoritative notation supplement makes the same normalization and recombination contract explicit in eq. 30 and eq. 31; those equations govern the symbols used by the implementation and all later supplements.

Conjugate likelihood learning for the shared model I

Active-inference agents learn the parameters of their generative model, not just plan with them (Smith et al. 2022; Friston et al. 2024). The likelihood matrix A carries a Dirichlet prior with concentration a over each column, updated conjugately by accumulating observation-state co-occurrence counts (eq. 14), giving the column-normalized expected likelihood. The update of eq. 14 is driven by the expected sufficient statistics under the data-generating model, so as the concentrations accumulate $\mathbb{E}[A]$ converges to the true likelihood. Convergence is measured by the per-column KL divergence summed over hidden states, which decreases monotonically toward the standard-Bayes fixed point; sec. 7.3 reports the learning curve, where the KL falls from 3.4231 to 0.0027 across 24 count batches. A forgetting hyperprior optionally decays the running mass toward an

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asymptote so the agent does not become infinitely confident; with the hyperprior disabled the classical unbounded accumulation of eq. 14 is recovered. The implementation is `dirichlet_learning.learn_likelihood`.

Bayesian model reduction for structure comparison I

Structure learning in the active-inference frame proceeds by Bayesian model reduction (BMR): given a full model with Dirichlet posterior `post` under prior `prior`, the change in negative variational free energy from swapping in a *reduced* prior — for example one that prunes a redundant column toward zero — is available in closed form without re-running inference (Friston and Penny 2011; Smith et al. 2022). Because the likelihood is shared, the reduced posterior is `post + reduced_prior - prior`, and the free-energy difference is a difference of log multivariate Beta functions (eq. 16), where $\ln B(a) = \sum_k \ln \Gamma(a_k) - \ln \Gamma(\sum_k a_k)$ is the log Dirichlet normalizer. A positive ΔF in eq. 16 means the reduced model has more evidence — the pruned structure was redundant and should be adopted; a negative ΔF means the

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reduction destroyed something the data support. When the reduced prior equals the prior the score is identically zero, the no-reduction fixed point. sec. 7.4 reports $\Delta F = 3.68$ for a redundant reduction (accepted) against $\Delta F = -27.67$ for a supported one (rejected). The implementation is `bayesian_model_reduction.reduce`.